



Convex Optimization

Project 4



Spring 1403
Due date: 28th of Ordibehesht

1. *Minimax, maximin, and minimum average error detectors.* We consider the optimal detector design problem as in §7.3 of the book. Suppose X is a random variable with values in $\{1, \dots, n\}$, with a distribution that depends on a parameter $\theta \in \{1, \dots, m\}$. The m possible distributions of X are represented by a matrix $P \in \mathbf{R}^{n \times m}$, with the j th column of P representing the distribution of X when $\theta = j$. Our goal is to design a (possibly) randomized detector, given by a matrix $T \in \mathbf{R}^{m \times n}$, with entries

$$t_{ik} = \text{prob}(\hat{\theta} = i \mid X = k), \quad i = 1, \dots, m, \quad k = 1, \dots, n$$

Evidently, we must have $t_{ik} \geq 0$ and $\sum_i t_{ik} = 1$. Note that $\hat{\theta}$ represents an estimated value of θ , as in statistics convention.

We let $D = TP \in \mathbf{R}^{m \times m}$ denote the detection probability matrix. Its off-diagonal entries $D_{ij}, i \neq j$ are the error probabilities. The diagonal entry D_{ii} is the accuracy of detecting $\theta = i$, $D_{ii} = \text{prob}(\hat{\theta} = i \mid \theta = i)$.

The following problems concern the specific problem instance with probabilities given in the `minimax_detector_data.py`. This file includes code that plots the m different distributions, as well as a detector matrix T (which you will solve for) as a stack plot. We consider three different detector design problems.

- (a) *Minimax detector.* Minimize the maximum error probability $\max_{i \neq j} D_{ij}$.
- (b) *Maximin detector.* Maximize the minimum detection accuracy $\min_i D_{ii}$.
- (c) *Minimum average error detector.* Minimize the average error $\frac{1}{m^2 - m} \sum_{i \neq j} D_{ij}$.

For each case, give the optimal value. Also give an optimal detection probability matrix D^* , and a stack plot of an optimal detector T^* .

Remarks.

- The problems above need not have unique solutions.
- Be sure to sanity check your results. Generally speaking, detectors should assign high probability for a given outcome to distributions that have high probability for that outcome.

Good Luck!